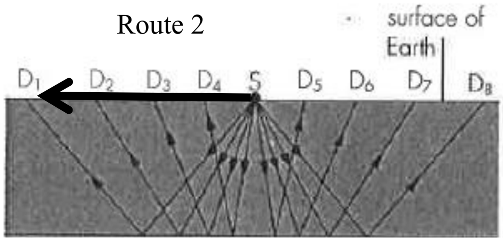
8 **(a)** A longitudinal wave is one in which the <u>oscillation of the molecules</u> of the wave is <u>along the direction of transfer of energy</u> of the wave. **A1**

(b) $A = \rho v^2 = (2700)(3100)^2 = 2.59 \times 10^{10}$ **A1**

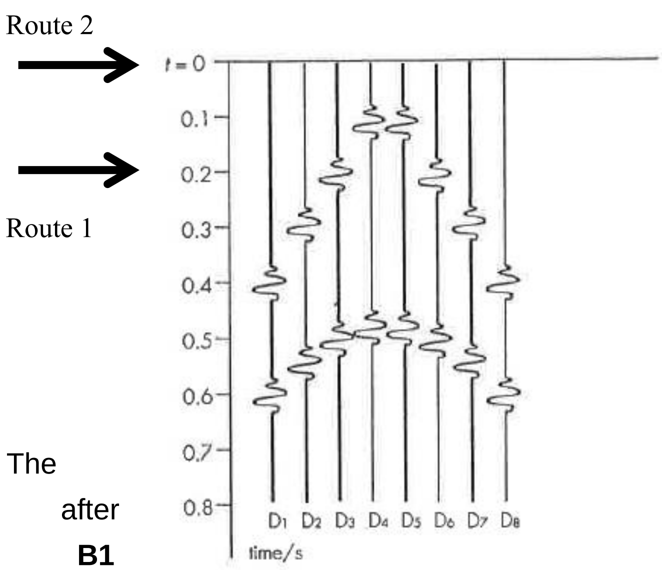
Units of A = $(\text{kg m}^{-3})(\text{m s}^{-1})^2 = \text{kg m}^{-1}\ \text{s}^{-2}$ = Pa **A1**

(c)



A1

(d)



A1

(e) The waves should be weaker after traveling longer distances, **B1** hence direct waves should show larger amplitude than reflected waves. **B1**

(f) **(i)** SD_8, t = 0.40 s

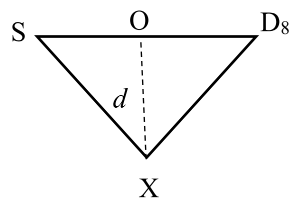
SD_8 = (3.1)(0.40) = 1.24 km **A1**

(ii) SXD_8, t = 0.60 s

SXD_8 = (3.1)(0.60) = 1.86 km **A1**

(g) Assume $SX = XD_8$

Using Pythagoras Theorem,



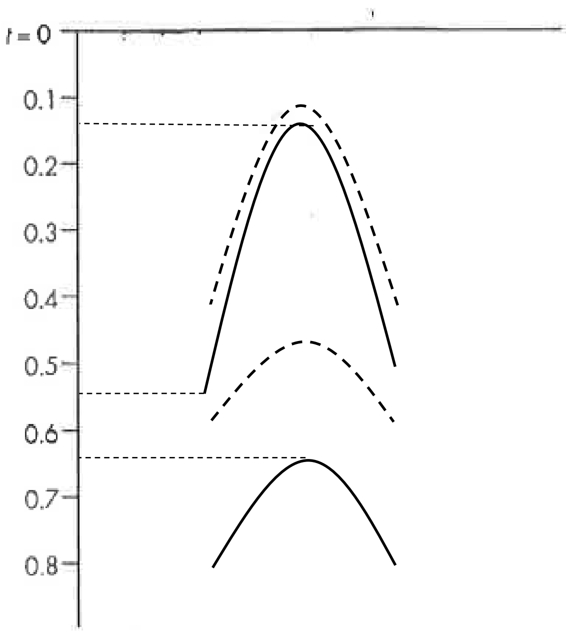
$$\text{depth d} = \sqrt{(XD_8)^2 - (OD_8)^2}$$

$$= \sqrt{\left(\frac{SXD_8}{2}\right)^2 - \left(\frac{SD_8}{2}\right)^2}$$ **M1**

$$= \sqrt{\left(\frac{1.89}{2}\right)^2 - \left(\frac{1.27}{2}\right)^2}$$

= 0.70 km **A1**

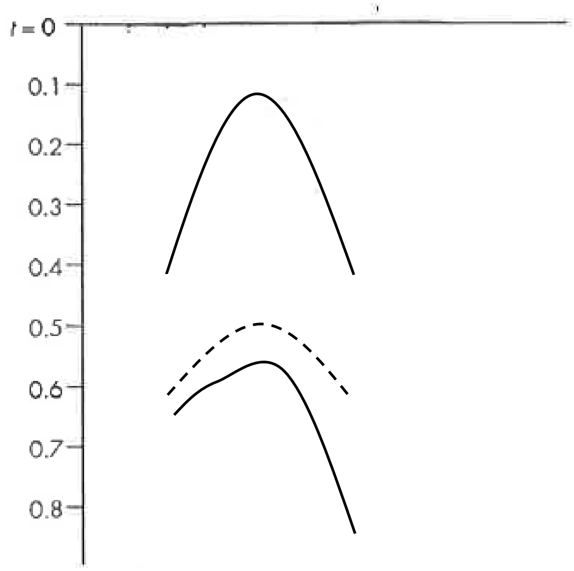
(h) (i)



Graph is lower, **B1**

Any peak value at a time factor about (3.1/2.4) = 1.3 times later **B1**

(ii)



Top graph is same, **B1**

Lower graph is lower and asymmetric as shown **B1**

(iii) Any one of the following:

- An extra layer of rock halfway down that can cause partial reflection
- Double reflection before reaching detector
- Some refraction takes place at intermediate level

(as a result of density changes) **B1**

- THE END -